

Basic OOP Principles Theory and Reasoning

Instructions: Answer the following questions in your own words. Explain your reasoning when a situation is presented.

1. **Why are Encapsulation, Inheritance, Polymorphism, and Abstraction considered important principles of OOP? Explain in your own words**

These four principles are fundamental because they allow developers to create software that is organized, secure, reusable, and easy to maintain. They help model real-world problems by breaking them down into logical pieces of code (objects), avoiding code repetition, and reducing the overall complexity of systems as they grow.

2. **A program has several types of employees. All employees have a name and an ID, but managers and programmers have different responsibilities. Which OOP principle could you use to represent this relationship? Explain why.**

The appropriate principle here is Inheritance. You can create a parent class called Employee; that contains the shared attributes like name and ID. Then, you can create child classes like Manager and Programmer that inherit from Employee. This way, you reuse the code for common attributes and only program the specific responsibilities of each role in their respective child classes.

3. **A system has different types of vehicles, such as cars, motorcycles, and trucks. They can all perform a common action, but each type performs it differently. Which OOP principle would be useful in this situation? Explain your answer.**

The principle to use is Polymorphism. Polymorphism allows different objects to share the same method name for an action (e.g., `accelerate()` or `brake()`), but each vehicle executes that action in its own specific way. Thus, the system can instruct any vehicle to perform the common action without needing to know exactly what type of vehicle it is.

4. **A class contains information that should not be directly accessible or modified from outside the class. Which OOP principle would help protect this information? Explain why.**

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The principle that helps protect this information is Encapsulation. Encapsulation restricts direct access to an object's internal data. By making this information private, you ensure that nothing outside the class can modify it directly. Instead, others are forced to use specific methods (like getters and setters) provided by the class to control and validate any changes.

5. **A software system contains many complex operations, but the user only needs to interact with a few simple features. Which OOP principle could help hide the unnecessary complexity? Explain your reasoning.**

This is a clear example of Abstraction. Abstraction involves exposing only the essential features and functions of an object to the user, hiding all the complex details of its internal workings. This makes the system easier to use, as the user only interacts with a simple interface rather than the complex underlying operations.

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